Conditional Inference in Two-Stage Response-Adaptive Experiments Via the Bootstrap

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Abstract Two-stage response-adaptive designs update the initial estimate of model parameters using data accumulated in the first stage. This updated estimate is then used to select the design matrix for a second stage. Inference following such procedures is complicated by the non-ancillarity of the design. To avoid this issue, inference is often conducted using either the observed or expected information derived as if the design were ancillary and ignoring its dependence on the earlier stage responses. We find this can provide poor inference. To account for these complications we develop bootstrap methods motivated by conditional inference.

1 The Model and an Illustration of the Problem

In many experiments the primary design objective is defined as a function, $\psi(\cdot)$, of the model parameters. Examples include optimal designs of nonlinear functions of the parameters in linear models; linear functions of parameters in nonlinear models; dose-finding studies where patients are treated at the estimate of a pre-specified quantile of the response function; and interim analysis where an experiment can be terminated early to reduce expected cost. In each experimental setting it is appealing to update initial estimates using accumulated data and use them to select the design matrix for a second stage. This procedure results in a design matrix that is no longer

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ancillary. In this section we examine the potential bias and difficulty in approximating the variance of maximum likelihood estimates following an adaptive design. In Section 2 three bootstrap methods are developed to obtain more accurate inference.

Consider an experiment conducted to measure a constant θ. Suppose it is possible to obtain unbiased observations of θ from two sources; each source $k = r,s$ produces errors from a $\mathcal{N}(0,\sigma_k^2)$, where $\sigma_r^2 \neq \sigma_s^2$ are known constants. A function $\psi(\theta)$ is defined to determine which of the two sources should be used based on maximizing efficiency, ethics, cost or other considerations. For illustration we use $\psi(\theta) = r$ if $\theta < c$ and s otherwise.

The adaptive procedure is as follows; obtain an initial cohort of n_1 observations from source r. Evaluate the function ψ at the maximum likelihood estimate (MLE) of the mean θ based on first stage observations, $\hat{\theta}_1 = \bar{y}_1$. The value $\psi(\bar{y}_1)$ determines the source from which a second cohort of n_2 observations will be obtained. This model was used in [1] and [3] to illustrate the need for conditional inference in experiments with ancillary designs. Here we use the model for the purpose of illustrating the need for and difficulty of conditional inference in adaptive experiments.

Throughout this paper $Y_i = (Y_{i1},\ldots,Y_{in_i})$ and $y_i = (y_{i1},\ldots,y_{in_i})$ denote the random sample from the ith stage and its observed realization, respectively. Let $N = n_1 + n_2$ be the total sample size; and, for notational convenience, let $S_k = \{\theta \in \mathbb{R} : \psi(\theta) = k\}$ be the region of the parameter space that selects the second stage source $k = r$ or s, eg., $S_r = (-\infty, c)$.

Consider the log of the likelihood $l_\theta \propto n_1(\bar{y}_1 - \theta)^2/2\sigma_r^2 + n_2(\bar{y}_2 - \theta)^2/2\sigma^2_{\psi(\bar{y}_1)}$. Let $w = (n_1/\sigma_r^2 + n_2/\sigma^2_{\psi(\bar{y}_1)})^{-1}$. Then the MLE based on both stages is $\hat{\theta} = w(n_1\bar{y}_1/\sigma_r^2 + n_2\bar{y}_2/\sigma^2_{\psi(\bar{y}_1)})$. If the design were ancillary, then the observed and expected information matrices are $\mathscr{I}(\overline{Y}_1) = -[\partial^2 l_\theta/\partial\theta^2]_{\theta=\hat{\theta}} = \mathrm{Var}\left[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|\overline{Y}_1)\right]$ and $\mathscr{F}_\theta = \mathrm{E}_{\overline{Y}_1}\left[\mathscr{I}(\overline{Y}_1)\right]$. In inference with ancillary designs, the observed information is considered a conditional measure, whereas expected information is unconditional. Here we want the variance conditioned on the assignment of the second stage design, which is $\sum_k \mathrm{Var}\left[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|\overline{Y}_1 \in S_k\right] I(\overline{Y}_1 \in S_k)$, where $I(\cdot)$ is the indicator function. When the meaning is clear we use S_k in place of $\overline{Y}_1 \in S_k$ for shorthand. See Table 1 columns 4 and 5 for a comparison of these. Values of $\theta = 1$, $\sigma_r = 1$, $\sigma_s = 3$, $n_1 = 50$, $n_2 = 150$ and $c = \theta + 1/\sqrt{n_1}$ are used throughout. Both the observed and expected information are significantly greater than the true conditional variance, indicating that the use of either as an approximation to the conditional variance is problematic. The conditional variances are also significantly less than the marginal variance, $\mathrm{Var}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)]$.

There doesn't exist in the literature a conclusive recommendation for inference in adaptive experiments. It is common to use either the observed or expected information and to create confidence intervals using normal quantiles. In this example the distribution of $\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_k$, $k = r,s$ can be found exactly. However, even in this simple case the conditional bias and variances of the MLE depend on the value of the parameter, i.e., they are not pivotal. Even in slightly more complex examples, derivation of the conditional distributions is likely not possible. For the purpose of this illustration, values of the expected value, variance, observed and expected

information, along with tail probabilities using these values, are given. In general the only value that is available is the observed information. The others must be approximated. Therefore these values should be , viewed as benchmark or comparison values and not alternatives for inference.

Table 1 presents $P(\overline{Y}_1 \in S_k)$, $\mathrm{E}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_k]$, $k = r,s$, $N\widehat{se}^2$, where $\widehat{se}^2$ is either the true variance, observed or expected information, and tail probabilities $\mathrm{P}[\theta < C_l]$ and $\mathrm{P}[\theta > C_u]$, where $C_l = \hat{\theta} - Z_{1-\alpha/2}\widehat{se}$ and $C_u = \hat{\theta} + Z_{1-\alpha/2}\widehat{se}$, respectively, and Z_α is the α quantile of the standard normal distribution. Nominally $\alpha = 0.05$. If $\psi(\bar{y}_1) = r$, then the mean is nearly unbiased; and despite the unbalanced tail probabilities; overall coverage would be adequate if the true variance were known and could be used. However, when the variance is approximated, using either observed or expected information, resulting intervals are too wide. When $\psi(\bar{y}_1) = s$, the bias is considerable and the coverage is unacceptable. Due to the poor performance when $\psi(\bar{y}_1) = s$ we will highlight improvements for this case throughout.

2 Conditional Bootstrap Inference

In an effort to reduce the negative effects of each, three bootstrap methods are developed. The first bootstrap procedure is a straightforward and intuitive conditional bootstrap procedure. Unfortunately, as will be discussed, the conditions required for this procedure to give accurate inference are extremely restrictive. Let * denote the bootstrap samples.

Conditional Bootstrap Method 1 (BM1)

1. Construct a probability distribution, $\hat{F}_i$, by putting mass $1/n_i$ at each point $y_{i1},\ldots,y_{in_i}$, $i = 1,2$. With fixed $\hat{F}_1$ and $\hat{F}_2$ draw a random sample of size n_1 and n_2 from $\hat{F}_1$ and $\hat{F}_2$, respectively. Denote these samples $y_i^*, i = 1,2$.
2. Repeat step 1 B times. For the bootstrap samples satisfying $\overline{y}_1^* \in S_{\psi(\bar{y}_1)}$, use y_1^* and y_2^* to find $\hat{\theta}^*$. Use $(C_l,C_u) = (Q^*_{\alpha/2}, Q^*_{1-\alpha/2})$ as the $(1-\alpha)$ confidence interval, where Q^*_α is the α quantile of the bootstrapped sample distribution $\hat{\theta}^*_{\psi(\bar{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*) = \hat{\theta}^*(\overline{Y}_1^*,\overline{Y}_2^*)|S_{\psi(\bar{y}_1)}$.

Let P^* be the empirical probability measure given Y_1 and $[\cdot]_b$ represent the bth sampled bootstrap. Then $P^*(\overline{Y}_1^* \in S_{\psi(\bar{y}_1)}) \approx \sum_{b=1}^B I([\overline{y}_1^*]_b \in S_{\psi(\bar{y}_1)})/B$ is the probability a given bootstrap sample is within the $S_{\psi(\bar{y}_1)}$.

For the cases in which $\psi(\bar{y}_1) = s$, the $P^*(\overline{Y}_1^* \in S_s)$, $\mathrm{E}^*\left[\hat{\theta}_z^*\right]$ and $\mathrm{Var}^*\left[\hat{\theta}_z^*\right]$ are presented in the fourth row of Table 2 using $B = 1000$ bootstrap samples. The first row presents the values for the true distribution of $\hat{\theta}$; the second and third row give results using the observed and expected information evaluated at $\hat{\theta}$. The confidence limits provided by each method are given as well as tail probabilities. All values are averages from 10,000 simulations except $P^*(\overline{Y}_1^* \in S_s)$ and $P(\overline{Y}_1 \in S_s|\theta = \hat{\theta})$, which are very skewed and for which the median provides a better summary of the

Table 1 Conditional probability, expectation, the method by which the variance was approximated (true variance, observed and expected information) along with its value and simulated $\alpha = 0.05$ in the regions above and below the confidence limits $C_l = \hat{\theta} - Z_{1-\alpha/2}\widehat{se}$ and $C_u = \hat{\theta} + Z_{1-\alpha/2}\widehat{se}$.

k	$P(\overline{Y}_1 \in S_k)$	$\mathrm{E}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)\|S_k]$	Method	$N\widehat{se}^2$	$\mathrm{P}[\theta < C_l]$	$\mathrm{P}[\theta > C_u]$
r	0.77	0.99	$\mathrm{Var}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)\|S_{\psi(\overline{Y}_1)}]$	0.88	0.02	0.04
			$\mathscr{I}(\bar{y}_1)$	1.00	0.01	0.03
			$\mathscr{F}_\theta$	1.47	0.00	0.01
			$\mathrm{Var}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)]$	1.77	0.00	0.01
s	0.23	1.14	$\mathrm{Var}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)\|S_{\psi(\overline{Y}_1)}]$	1.24	0.39	0.0
			$\mathscr{I}(\bar{y}_1)$	3.00	0.10	0.00
			$\mathscr{F}_\theta$	1.47	0.33	0.00
			$\mathrm{Var}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)]$	1.77	0.26	0.00

data. Both approximate probabilities are significantly greater than $P(\overline{Y}_1 \in S_s)$. These approximations contribute to the overestimation of the variance using BM1 and $\mathscr{F}_{\hat{\theta}}$. The observed information also significantly over estimates the variance. The mean of the bootstrap distribution has greater bias than $\hat{\theta}$. In this example use of BM1 certainly is not an improvement over $\mathscr{F}_{\hat{\theta}}$.

To examine the poor performance in more detail consider $\overline{Y}_1^*|\overline{Y}_1^* \in S_k = (a_k, d_k)$, where $a_k = \theta + a_k'/\sqrt{n_1}$, $d_k = \theta + d_k'/\sqrt{n_1}$, a_k' and d_k' are constants. This set-up is a slightly more general representation of S_k than is given by the previous definition of ψ and assumes that the initial estimate of θ and the definition of ψ are such that the interval $P(\overline{Y}_1 \in S_k)$ has positive probability not equal 1. If $P(\overline{Y}_1 \in S_k) = 1$ for some k then the design would be determinant, not adapted. Now

$$\begin{aligned}
&P^*\left(\sqrt{n_1}\left(\overline{Y}_1^* - \overline{Y}_1\right) < x | \sqrt{n_1}\left(\overline{Y}_1^* - \overline{Y}_1\right) \in \sqrt{n_1}(a_k - \theta, d_k - \theta)\right) \\
&\quad = \frac{P^*\left(\left\{\sqrt{n_1}\left(\overline{Y}_1^* - \overline{Y}_1\right) < x\right\} \cap \left\{\sqrt{n_1}(\overline{Y}_1^* - \overline{Y}_1) \in \sqrt{n_1}(a_k - \theta, d_k - \theta)\right\}\right)}{P^*\left(\sqrt{n_1}(\overline{Y}_1^* - \overline{Y}_1) \in \sqrt{n_1}(a_k - \theta, d_k - \theta)\right)} \\
&\quad = \frac{P\left(\left\{\sqrt{n_1}\left(\overline{Y}_1 - \theta\right) < x\right\} \cap \left\{\sqrt{n_1}(\overline{Y}_1 - \theta) \in \sqrt{n_1}(a_k - \theta, d_k - \theta)\right\}\right) + o(1)}{P\left(\sqrt{n_1}(\overline{Y}_1 - \theta) \in \sqrt{n_1}(a_k - \theta, d_k - \theta)\right) + o(1)} \\
&\quad = P\left(\sqrt{n_1}\left(\overline{Y}_1 - \theta\right) < x | \sqrt{n_1}\left(\overline{Y}_1 - \theta\right) \in \sqrt{n_1}(a_k - \theta, d_k - \theta)\right) + o(1), \qquad (1)
\end{aligned}$$

where $x \in \mathbb{R}$, and the equalities hold almost surely under P. Unfortunately, θ is unknown and must be estimated. In order to replace θ with an estimate, say $\tilde{\theta}$, in (1) $\tilde{\theta} \to \theta$ at a rate faster than $1/\sqrt{n_1}$, i.e.,

$$\begin{aligned}
&P^*\left(\sqrt{n_1}\left(\overline{Y}_1^* - \overline{Y}_1\right) < x | \sqrt{n_1}\left(\overline{Y}_1^* - \overline{Y}_1\right) \in \sqrt{n_1}(a_k - \tilde{\theta}, d_k - \tilde{\theta})\right) \\
&\quad = P\left(\sqrt{n_1}\left(\overline{Y}_1 - \theta\right) < x | \sqrt{n_1}\left(\overline{Y}_1 - \theta\right) \in \sqrt{n_1}(a_k - \theta, d_k - \theta)\right) + o_P(1). \qquad (2)
\end{aligned}$$

In a single stage experiment this would not be possible. However, in a two stage adaptive experiment if $n_1 = o(n_2)$, then such an estimate may exist. Theoretically this is somewhat unsatisfying, but practically this is interesting especially since [4]

and [5] show that two stage experiments are optimized when $n_1 = O(\sqrt{n_2})$. It is also common, for practical or logistical reasons, that a small pilot study is precedes a much larger follow-up.

Estimate	Method	$P(\overline{Y}_1 \in S_s)$	$\mathrm{E}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)\|S_s]$	N*Variance	C_l	C_u	$\mathrm{P}[\theta < C_l]$	$\mathrm{P}[\theta > C_u]$
$\hat{\theta}$	True	0.23	1.14	1.24	0.99	1.30	-	-
$\hat{\theta}$	$\mathscr{F}_{\hat{\theta}}$	0.59	1.14	1.82	0.95	1.32	0.30	0.00
$\hat{\theta}$	$\mathscr{I}(\bar{y}_1)$	-	1.14	3.00	0.90	1.38	0.10	0.00
$\hat{\theta}^*$	BM1	0.67	1.19	1.87	1.02	1.39	0.63	0.00
$\hat{\theta}^+$	BM2	0.25	1.13	1.36	0.99	1.30	0.41	0.00
$\tilde{\theta}$	True	0.23	1.00	5.25	0.66	1.28	-	-
$\tilde{\theta}$	$\mathscr{F}_{\tilde{\theta}}$	0.32	1.00	2.35	0.79	1.21	0.14	0.16
$\tilde{\theta}^+$	BM3	0.25	0.98	5.60	0.65	1.29	0.02	0.02

Table 2 Results are conditional on the region $\overline{Y}_1 \in S_s$. Method describes the procedure by which the approximate distribution was obtained. $P(\overline{Y}_1 \in S_s)$ is approximated by $\sum_{b=1}^{B} I([\overline{y}_1^*]_b \in S_{\psi(\bar{y}_1)})/B$ for the bootstrap and $P(\overline{Y}_1 \in S_s|\theta = \hat{\theta})$ for expected information. $\mathrm{E}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s)]$ and the variance, the confidence limits and tail probabilities are also provided. All values are averages from 10,000 simulations except the column $P(\overline{Y}_1 \in S_s)$ which uses the median.

Note $\overline{Y}_1|\overline{Y}_1 \in S_k \in\sim \mathscr{T}\mathscr{N}(\theta,\sigma_1^2/n_1;\overline{Y}_1 \in S_k)$, where $\mathscr{T}\mathscr{N}$ denotes a truncated normal distribution. From (2), if an estimate exists as described, then $\overline{Y}_1^*|\{\overline{Y}_1^* \in S_k\} - \bar{y}_1$ has the same approximate distribution as $\overline{Y}_1|\{\overline{Y}_1 \in S_k\} - \mathrm{E}[\overline{Y}_1|\overline{Y}_1 \in S_k]$. This implies that $\overline{Y}_1^*|\overline{Y}_1^* \in S_k$ can be approximated by a $\mathscr{T}\mathscr{N}(\bar{y}_1,\sigma_1^2/n_1;\overline{Y}_1^* \in \psi(S_k))$.

Now to refine the bootstrap, observe that the truncation results in an additive bias

$$\mathrm{E}\left[\overline{Y}_1|\overline{Y}_1 \in S_k\right] = \theta + b_{1k}(\theta),$$

where $b_{1k}(\theta) = (\phi(\gamma(a_k)) - (\phi(\gamma(d_k))/U$, $\gamma(a) = \sqrt{n_1}(a-\theta)/\sigma_1$, $U = \Phi(\gamma(b)) - \Phi(\gamma(a))$; $\phi(\cdot)$ and $\Phi(\cdot)$ represent the PDF and cumulative distribution function (CDF) of the standard normal distribution, respectively. Let $\tilde{S}_k = (a_k - \tilde{\theta}, d_k - \tilde{\theta})$. Then using the approximate relationships we get

$$\mathrm{E}^*\left[\overline{Y}_1^*|\overline{Y}_1^* \in S_k\right] \approx \overline{Y}_1 + d_{1k}(\overline{Y}_1) \tag{3}$$

$$\mathrm{E}^*\left[\overline{Y}_1^*|\overline{Y}_1^* \in \tilde{S}_k\right] \approx \overline{Y}_1 + d_{1k}(\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)). \tag{4}$$

Similarly for the variance

$$\begin{aligned}\mathrm{Var}\left[\overline{Y}_1|\overline{Y}_1 \in S_k\right] &= V(a_k,d_k,\theta)\\ &= \frac{\sigma_1}{\sqrt{n_1}}\left[1 + \frac{\gamma(a_k)\phi(\gamma(a_k)) + \gamma(d_k)\phi(\gamma(d_k))}{U}\right] + (b_{1k}(\theta))^2;\end{aligned}$$

and for the bootstrap we get

$$\mathrm{Var}\left[\overline{Y}_1^*|\overline{Y}_1^* \in S_k\right] \approx V(a_k,d_k,\overline{Y}_1) \tag{5}$$

$$\mathrm{Var}\left[\overline{Y}_1^*|\overline{Y}_1^* \in \tilde{S}_k\right] \approx V(a_k,d_k,\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)). \tag{6}$$

Equations (3) and (5) show explicitly why the bias and variance worsened in the simulated BM1 example. However, we can see that the bias and variance terms in equations (4) and (6) now depend on $\tilde{\theta}$. If $\tilde{\theta}$ is an asymptotically unbiased estimate, then we can expect the bootstrap variance to be approximately equal to the variance of $\overline{Y}_1$; however it adds $b_{1k}(\tilde{\theta})$ to the expectation. This will have to be corrected in the bootstrap. We adapt the bias reduction method suggested by [6]. Note

$$\begin{aligned}\mathrm{E}\left[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|\overline{Y}_1 \in S_k\right] &= w\mathrm{E}\left[n_1\overline{Y}_1/\sigma_1^2 + n_2\overline{Y}_2/\sigma_2^2|\overline{Y}_1 \in S_k\right] \\ &= \theta + wn_1 b_{1k}(\theta)/\sigma_1^2.\end{aligned}$$

Let $b_k(\theta) = wn_1 b_{1k}(\theta)/\sigma_1^2$. Then a bias corrected estimate of θ is $\tilde{\theta} = \hat{\theta} - b_k(\tilde{\theta})$. To solve this equation the Newton-Raphson method was used. Typically one iteration was sufficient, i.e.,

$$b_k^{'}(\tilde{\theta}) = b_k(\hat{\theta})/(1_p + [\partial b_k(\theta)/\partial\theta]_{\theta=\hat{\theta}})^{-1},$$

where 1_p is a p dimensional vector of with each entry equal to 1. This formulation was used to for analytic expressions and numeric calculations. From this we develop the following bootstrap procedure

Adjusted Conditional Bootstrap Method (BM2): Repeat BM1 keeping only bootstrap samples satisfying $\overline{y}_1^* \in \tilde{S}_{\psi(\overline{y}_1)}$. Let $\hat{\theta}^+ = \hat{\theta}^* - b_{\psi(\overline{y}_1)}(\tilde{\theta})$ and $(C_l^+, C_u^+) = (Q_{\alpha/2}^+, Q_{1-\alpha/2}^+)$ as the $(1-\alpha)$ confidence interval, where Q_α^+ is the α quantile of the bootstrap sample distribution of $\hat{\theta}^+_{\psi(\overline{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*) = \hat{\theta}^+(\overline{Y}_1^*,\overline{Y}_2^*)|\overline{Y}_1^* \in \tilde{S}_{\psi(\overline{y}_1)}$.

The use of $\hat{\theta}^+$ in place of $\hat{\theta}^*$ is not a bias correction in any traditional sense. It is an adjustment for the additional bias due to the conditional bootstrap procedure. When (4) is approximately true using $\hat{\theta}^+$ in place of $\hat{\theta}^*$ only adds Monte Carlo error and not error associated with the variance of $b_{\psi(\overline{y}_1)}(\tilde{\theta})$.

Figure 1 (top left) presents histograms of the bootstrap sample distributions of $\hat{\theta}^+_{\psi(\overline{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*) - \hat{\theta}$ corresponding to the 0.025, 0.50 and 0.975 of quantiles (dotted, dashed and dot-dashed line) of $\hat{\theta}$ and a histogram of the distribution of $\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s - \mathrm{E}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s]$ (solid line). This figure illustrates how well BM2 works across the domain of $\hat{\theta}$ at approximating the target distribution. Compare this to Figure 1 (top right) which plots the PDF of $\mathscr{N}(0,\mathscr{F}_{\hat{\theta}}^{-1})$ for $\mathscr{F}_{\hat{\theta}}^{-1}$ evaluated at the same quantiles of $\hat{\theta}$. Here we can see that the variance of bootstrap distribution is less sensitive to the value of $\hat{\theta}$ than $\mathscr{F}_{\hat{\theta}}^{-1}$ and provides a better approximation to the shape of the target distribution.

In the fifth row of Table 2 the results for the same simulation are presented for BM2. Here we can see that $P^*(\overline{Y}_1^* \in \tilde{S}_s)$ is nearly equal to $P(\overline{Y}_1 \in S_s)$; $\mathrm{E}^*[\hat{\theta}^+_{\psi(\overline{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*)]$ is approximately equal to $\mathrm{E}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s]$; $\mathrm{Var}^*[\hat{\theta}^+_{\psi(\overline{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*)]$ is only slightly greater than $\mathrm{Var}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s]$; and the average confidence region of bootstrap correspond to the quantiles of $\hat{\theta}$. It is clear that BM2 provides a more accurate representation of the distribution of $\hat{\theta}$ than the alternatives. However, BM2 still fails

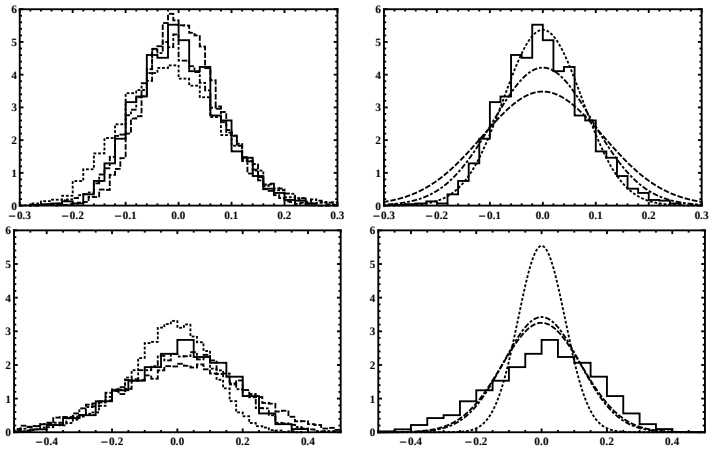


Fig. 1 Histogram of bootstrap distributions $\hat{\theta}^+(\overline{Y}_1^*,\overline{Y}_2^*)-\hat{\theta}$ (top left) and $\tilde{\theta}^+(\overline{Y}_1^*,\overline{Y}_2^*)-\tilde{\theta}$ (bottom left). Probability density functions of $N(0,\mathscr{I}_{\hat{\theta}}^{-1})$ (top right) and $N(0,\mathscr{I}_{\tilde{\theta}}^{-1})$ (bottom right). The dotted, dot dashed and dashed lines correspond to the bootstrap distribution or fisher information for the 0.025, 0.50, and 0.975 of quantiles of $\hat{\theta}$ or $\tilde{\theta}$. In each case the solid line is the histogram of the true target distribution of $\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s-\mathrm{E}[\hat{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s]$ or $\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s-\mathrm{E}[\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s]$.

to provide adequate coverage. This is not a problem with the bootstrap but rather a problem with the distribution of $\hat{\theta}$ being tightly centered around the wrong value.

Due to the poor coverage of BM2 it is of interest to derive a bootstrap procedure for $\tilde{\theta}$. Note as was previously mentioned for this simple example it is possible to derive the true conditional distributions of $\hat{\theta}$. The derivation of the conditional distribution of $\tilde{\theta}$ is much more difficult, however; we are still able to approximate its bootstrap sample distribution.

Bias Adjusted Conditional Bootstrap Method (BM3): Replace $\hat{\theta}^+$ in BM3 with $\tilde{\theta}^+=\hat{\theta}^+-b'_{\psi(\bar{y}_1)}(\hat{\theta}^+)$. Use $(C_l,C_u)=(\widehat{CDF}^{-1}(\Phi(2v_0+Z_{\alpha/2})),\widehat{CDF}^{-1}(\Phi(2v_0-Z_{\alpha/2})))$, where $\widehat{CDF}(t)=P^*(\tilde{\theta}^+_{\psi(\bar{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*)<t)$, $\tilde{\theta}^+_{\psi(\bar{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*)=\tilde{\theta}^+(\overline{Y}_1^*,\overline{Y}_2^*)|\overline{Y}_1^*\in\tilde{S}_{\psi(\bar{y}_1)}$ and $v_0=\Phi^{-1}\widehat{CDF}(\tilde{\theta})$.

Recall from BM2 that $\hat{\theta}^+_{\psi(\bar{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*)|S_{\psi(\bar{y}_1)}$ can be used to approximate the distribution of $\hat{\theta}(\overline{Y}_1,\overline{Y}_2)$. From this $b'_z(\hat{\theta}^+_{\psi(\bar{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*))-b'_{\psi(\bar{y}_1)}(\hat{\theta})$ will have the same approximate distribution as $b'_{\psi(\bar{y}_1)}(\hat{\theta}(\overline{Y}_1,\overline{Y}_2))|S_s-\mathrm{E}[b'_{\psi(\bar{y}_1)}(\hat{\theta}(\overline{Y}_1,\overline{Y}_2))|S_s]$. Therefore $\tilde{\theta}^+_{\psi(\bar{y}_1)}(\overline{Y}_1^*,\overline{Y}_2^*)-\tilde{\theta}$ will have the same approximate distribution as $\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s-\mathrm{E}[\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s]$. The distribution of the bias, $b'_{\psi(\bar{y}_1)}(\hat{\theta}^+)$, is fairly skewed; the bias corrected bootstrap confidence intervals (see [2]) were used to account for this.

Figure 1 (bottom left) plots a histogram of bootstrap sample distributions of $\tilde{\theta}_s^+(\overline{Y}_1^*,\overline{Y}_2^*)-\tilde{\theta}$ corresponding to the 0.025, 0.50 and 0.975 quantiles (dotted, dashed and dot-dashed line) of $\tilde{\theta}$ and a histogram of the distribution of $\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s-\mathrm{E}[\tilde{\theta}(\overline{Y}_1,\overline{Y}_2)|S_s]$ (solid line). For comparison Figure 1 (bottom right) plots the prob-

ability density functions of a $\mathcal{N}(0, \mathscr{F}_{\tilde{\theta}}^{-1})$ for $\mathscr{F}_{\tilde{\theta}}^{-1}$ evaluated at the same quantiles of $\tilde{\theta}$. Once again we see that the bootstrap distribution provides a very nice approximation of the target distribution across most of the domain of $\tilde{\theta}$. It also represents a significant improvement of the alternative method in terms of sensitivity to $\tilde{\theta}$.

In the eighth row of Table 2 the results for the same simulation are presented for BM3. The sixth row presents the values for the true distribution of $\tilde{\theta}$ and the seventh row gives results for the expected information evaluated at $\tilde{\theta}$. The bootstrap distribution $\tilde{\theta}^+$ provides an unbiased estimate of the mean. It slightly inflates the variance of the target distribution $\tilde{\theta}$. The performance of $\tilde{\theta}^+$ represents a significant improvement over the alternatives.

Table 3 Mean square error (MSE) for the true distribution of $\hat{\theta}$ and $\tilde{\theta}$ along with the bootstrap MSE of $\hat{\theta}^+$ or $\tilde{\theta}^+$ for the case when $\psi(\bar{y}_1) = s$.

Estimate	Method	MSE
$\hat{\theta}$	True	4.72
$\hat{\theta}^+$	BM2	4.64
$\tilde{\theta}$	True	5.26
$\tilde{\theta}^+$	BM3	5.68

The use of $\hat{\theta}^+$ or $\tilde{\theta}^+$ is simply the bias versus variance tradeoff. Table 3 compares the mean square error (MSE) for the true distribution of $\hat{\theta}$ and $\tilde{\theta}$ along with the bootstrap MSE of $\hat{\theta}^+$ and $\tilde{\theta}^+$ for the case when $\psi(\bar{y}_1) = s$. Despite the shortcomings of the procedure for $\hat{\theta}^+$ it still provides a lower MSE than $\tilde{\theta}^+$ in this example.

3 Discussion

In this paper we examined how to improve inference using conditional bootstrap procedures. For the purpose of illustration a simple model was investigated. The practical extension to more complex situations is likely to be challenging but straightforward. Theoretical extension represents a more significant hurdle. Both of these topics require further research.

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